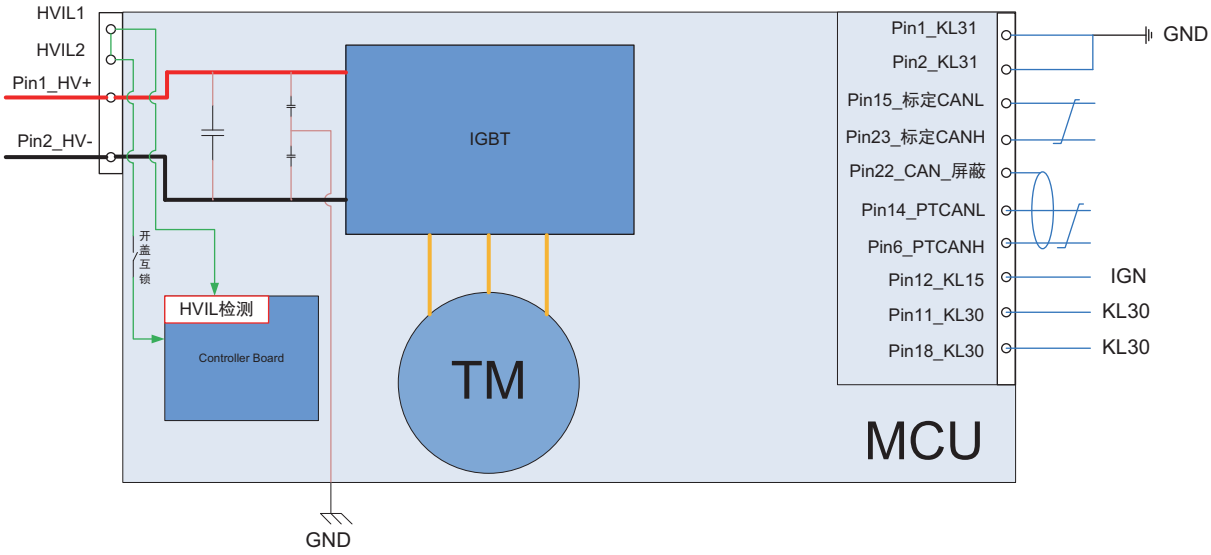
P190D-1C, P190E-1C、P190F-1C、P1910-1C、P1911-1C、P1912-1C、P1913-1C、P1914-1C、P1915-1C、 P1916-1C、 P1917-1C

Fault diagnosis code
P190D-1C：Power30P voltage over range
P190E-1C：V3P3 voltage out of range
P190F-1C：V15P_BB_Filt voltage out of range
P1910-1C：V5PSens voltage out of range
P1911-1C：V5P_BB voltage out of range
P1912-1C：V15P voltage over range
P1913-1C：-8V voltage over range
P1914-1C：V3P3_HV voltage is out of range
P1915-1C：V5P_HV voltage is out of range
P1916-1C：V12_EMsens voltage out of range
P1917-1C：V1P2 voltage out of range
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • PEU fault • High voltage insulation fault • The IGBT voltage circuit in the PEU is faulty
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P190D-1C
Power30P voltage is above the upper threshold （20V） or not below the lower threshold （5.5V）
P190E-1C
V3P3 voltage is above the upper threshold （3.45V） or not below the lower threshold （3.1V）
P190F-1C
V15P_BB_Filt voltage is above the upper threshold （15.75V） or not below the lower threshold （13.5V）
P1910-1C
V5PSens voltage is above the upper threshold （6V） or not below the lower threshold （4.75V）
P1911-1C
V5P_BB voltage is above the upper threshold （5.25V） or not below the lower threshold （4.75V）
P1912-1C

V15P voltage is above the upper threshold （18V） or not below the lower threshold （13V）
P1913-1C
-8V voltage is above the upper threshold （-7.2V） or not below the lower threshold （-8.8V）
P1914-1C
V3P3_HV voltage is above the upper threshold （3.45V） or not below the lower threshold （3.1V）
P1915-1C
V5P_HV voltage is above the upper threshold （5.25V） or not below the lower threshold （4.75V）
P1916-1C
V12_EMsens voltage is above the upper threshold （12.6V） or not below the lower threshold （11.4V）
P1917-1C
V1P2 voltage is above the upper threshold （1.26V） or not below the lower threshold （1.14V）
To set the effect of a fault code condition
When vehicle lost power, there is forced power at high speed

Circuit diagram



EV901001

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of PEU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the PEU with diagnostic apparatus. 4 If there is any fault code except for P190D-1C, P190E-1C, B12B4-12、P190F-1C、P1910-1C、P1911-1C、P1912-1C、P1913-1C、P1914-1C、P1916-1C、P1917-1C?
		YES → Refer to:DTC Summary list(PEU).
NO ↓		
Step 3	Check the connecting cable of PEU.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of PEU. 5 Check whether the PEU connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 4	Check the insulation of PEU.	
		1 Use a megohm meter to check the insulation resistance of the PEU housing and grounding according to the procedure in the maintenance manual. **Standard value: >20MΩ** 2 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.

YES ↓		
Step 5	Check the insulation of driving motor.	
		1 Use a megohm meter to check the insulation resistance of the driving motor housing and grounding according to the procedure in the maintenance manual. **Standard value: >20MΩ** 2 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 6	Perform initialization.	
		1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The governor was initialized with the diagnostic instrument. 5 Can the governor be initialized and is the governor restored?
		NO → Repair or change the governor.
YES ↓		
Step 7	Check PEU internal circuit.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check the PEU internal circuit and components for obvious damage or . ablation. 5 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 8	Test whether PEU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1002 of PEU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
	YES → Change the PEU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	